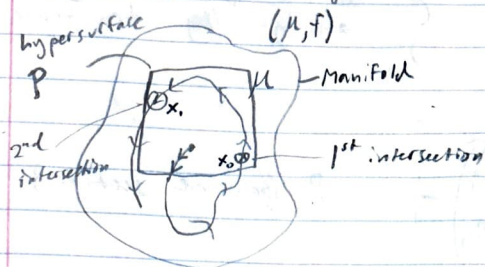


Lecture 10-21

Poincaré Sections

Dynamical system (general)



f : a system defined continuously in time

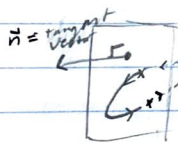


→ count intersections w/ hypersurface
→ converts from continuous time system to a discrete Mapping

$$x_{n+1} = IP(x_n)$$

→ For example, in celestial mechanics a Poincaré section is what allows us to evolve dynamical systems like the solar system forward in time while avoiding a lengthy integration

→ Consider a periodic orbit in a dynamical system



periodic orbits can be described in a single dimension

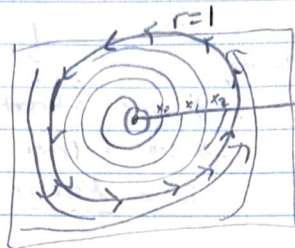
⇒ whenever you have continuous symmetry (i.e., the law doesn't change in time), the orbit can be described by a set of finite size

Example 8.7.1

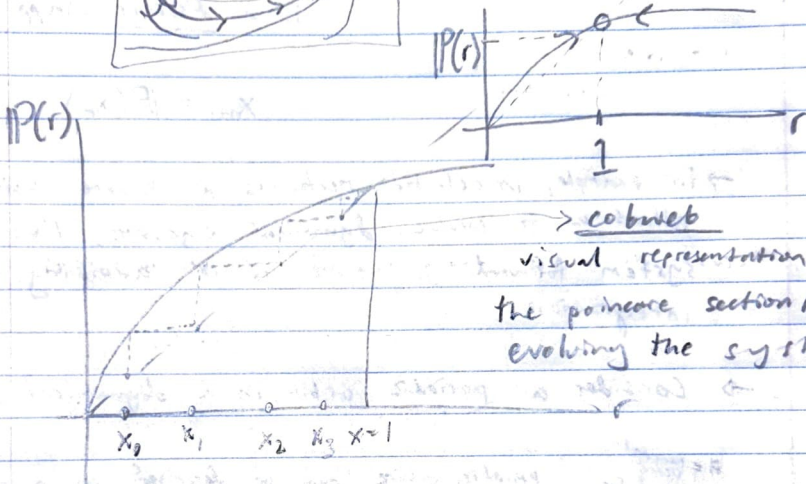
→ Consider the system:

$$\begin{cases} \dot{r} = r(1-r^2) \\ \dot{\theta} = 1 \end{cases}$$

simple system, only analytically tractable ex.



P (poinecare section)



→ cobweb
visual representation of
the poinecare section / map
evolving the system

to calculate analytically,

notice $2\pi = \int dt = \int \frac{dr}{r(1-r^2)}$

→ a note on notation:

$$\mathcal{J} = e^{\int \frac{1}{r} dr} \quad (\text{only in stogtze})$$

↑
Jacobian

Local Stability

→ Consider the autonomous dynamical system

$$\dot{x}_i = V_i(x)$$

$$\dot{x}_i + \delta \dot{x}_i = V_i(x + \delta x) = V_i(x) + \frac{\partial V_i}{\partial x_j} \delta x_j + \left(\frac{\partial^2 V_i}{\partial x_j^2} \delta x_j^2 + \dots \right) + \underbrace{O(\delta x^3)}_{\text{neglect higher order terms}}$$

$$\Rightarrow \dot{x}_i + \delta \dot{x}_i = V_i(x) + \frac{\partial V_i}{\partial x_j} \delta x_j + \dots$$

but recall

$$\dot{x}_i = V_i(x)$$

$$\Rightarrow \delta \dot{x}_i \approx \frac{\partial V_i}{\partial x_j} \delta x_j = \sum_j^N \frac{\partial V_i}{\partial x_j} \delta x_j$$

Recall the Poincaré map, which returns the trajectory to the next (in time) intersection of the orbit with a constructed hypersurface in phase space

$$x_{n+1} = IP(x_n)$$

$$= x_n + \int_{t_n}^{t_{n+1}} dt \dot{V}(x(t))$$

$$\Rightarrow x_{n+1} + \delta x_n = IP(x_n + \delta x_n) = IP(x_n) + \frac{\partial IP(x_n)}{\partial x} \delta x_n$$

and since $x_{n+1} = IP(x_n)$, we have shown

$$\delta x_{n+1} = \underbrace{\frac{\partial IP_i(x)}{\partial x_j}}_{L_n} \delta x_{n,j}$$

→ the matrix is taking ratios of distances: $\partial IP_i(x) / \partial x_j$
 $\Rightarrow$ elements are dimensionless

$$\Rightarrow \delta x_{n+1} = J(x_n) \delta x_n$$

(Jacobian)

$$J = \underbrace{\begin{Bmatrix} u_1, u_2, \dots, u_d \\ \Lambda_1, \Lambda_2, \Lambda_3, \dots \end{Bmatrix}}_{\text{invariant}} = \frac{\partial P(x_n)}{\partial x_i}$$

"under a very huge group" i.e., under all smooth coordinate transformations, these numbers will stay the same

This is all predicated on the existence of a periodic orbit

Poincaré sections: how do you compare things that look different in different coordinate systems

Chaos

Lorenz (1963) $\rightarrow$ Dame Cartwright (1945)

$\rightarrow$ Using Fourier mode symmetries and other analysis, Lorenz reduced a simplified format of NS to the following 3D system

$$\begin{cases} \dot{x} = \sigma(y - x) \\ \dot{y} = rx - y - xz \\ \dot{z} = -xy - bz \end{cases} \quad \sigma, r, b \in \mathbb{R}$$

equilibria? @ origin, but others unpredictable...